

Identity Mappings in Deep Residual Networks

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Abstract Deep residual networks [1] have emerged as a family of extremely deep architectures showing compelling accuracy and nice convergence behaviors. In this paper, we analyze the propagation formulations behind the residual building blocks, which suggest that the forward and backward signals can be directly propagated from one block to any other block, when using identity mappings as the skip connections and after-addition activation. A series of ablation experiments support the importance of these identity mappings. This motivates us to propose a new residual unit, which makes training easier and improves generalization. We report improved results using a 1001-layer ResNet on CIFAR-10 (4.62% error) and CIFAR-100, and a 200-layer ResNet on ImageNet. Code is available at: <https://github.com/KaimingHe/resnet-1k-layers>.

1 Introduction

Deep residual networks (ResNets) [1] consist of many stacked “Residual Units”. Each unit (Fig. 1 (a)) can be expressed in a general form:

$$\begin{aligned} \mathbf{y}_l &= h(\mathbf{x}_l) + \mathcal{F}(\mathbf{x}_l, \mathcal{W}_l), \\ \mathbf{x}_{l+1} &= f(\mathbf{y}_l), \end{aligned}$$

where $\mathbf{x}_l$ and $\mathbf{x}_{l+1}$ are input and output of the l -th unit, and $\mathcal{F}$ is a residual function. In [1], $h(\mathbf{x}_l) = \mathbf{x}_l$ is an identity mapping and f is a ReLU [2] function.

ResNets that are over 100-layer deep have shown state-of-the-art accuracy for several challenging recognition tasks on ImageNet [3] and MS COCO [4] competitions. The central idea of ResNets is to learn the additive residual function $\mathcal{F}$ with respect to $h(\mathbf{x}_l)$, with a key choice of using an identity mapping $h(\mathbf{x}_l) = \mathbf{x}_l$. This is realized by attaching an identity skip connection (“shortcut”).

In this paper, we analyze deep residual networks by focusing on creating a “direct” path for propagating information — not only within a residual unit, but through the entire network. Our derivations reveal that *if both $h(\mathbf{x}_l)$ and $f(\mathbf{y}_l)$ are identity mappings*, the signal could be *directly* propagated from one unit to any other units, in both forward and backward passes. Our experiments empirically show that training in general becomes easier when the architecture is closer to the above two conditions.

To understand the role of skip connections, we analyze and compare various types of $h(\mathbf{x}_l)$. We find that the identity mapping $h(\mathbf{x}_l) = \mathbf{x}_l$ chosen in [1]

深度残差网络中的恒等映射

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摘要 深度残差网络 [1] 作为一种极为深度的架构体系，这类网络展现出极高的准确率以及良好的收敛特性。在本文中，我们分析了残差构建模块背后的传播公式，这些公式表明：当使用恒等映射作为跳过连接，并结合加法后的激活函数时，前向信号和反向信号可以直接从一个模块传递到任意其他模块。一系列消融实验进一步印证了这些恒等映射的重要性。正因如此，我们提出了一种新的残差单元，它不仅能够简化训练流程，还能提升模型的泛化能力。我们报告了在CIFAR-10（误差率为4.62%）和CIFAR-100上使用含1001层的ResNet，以及在ImageNet上使用含200层的ResNet所取得的优化结果。相关代码可访问：

<https://github.com/KaimingHe/resnet-1k-layers>。

1 引言

D深度残差网络（ResNets） [1] 由众多堆叠在一起的“残差单元”构成。
E每个单元（见图1(a)）都可以用如下通用形式表示：

$$\begin{aligned} \mathbf{y}_l &= h(\mathbf{x}_l) + \mathcal{F}(\mathbf{x}_l, \mathcal{W}_l), \\ \mathbf{x}_{l+1} &= f(\mathbf{y}_l), \end{aligned}$$

其中 $\mathbf{x}_l$ 与 $\mathbf{x}_{l+1}$ 分别是第 l 个单元的输入和输出，而 $\mathcal{F}$ 则为残差函数。在 [1] 中， $h(\mathbf{x}_l) = \mathbf{x}_l$ 属于恒等映射， f 则是ReLU [2] 函数。

深度超过100层的ResNet网络在ImageNet [3] 及MS COCO [4] 竞赛中的多项高难度识别任务上，均展现出了处于行业领先水平的准确率。这类网络的核心思想是学习关于 $h(\mathbf{x}_l)$ 的加性残差函数 $\mathcal{F}$ ，其关键设计在于采用恒等映射 $h(\mathbf{x}_l) = \mathbf{x}_l$ 。这一设计是通过引入恒等跳连接（即“快捷路径”）来实现的。

在本文中，我们通过研究为信息传播构建“直接”路径的方式，对深度残差网络展开了分析——这种路径不仅存在于单个残差单元内部，也贯穿整个网络结构。我们的推导表明，如果 $h(\mathbf{x}_l)$ 以及 $f(\mathbf{y}_l)$ 都是恒等映射，那么无论是在正向传播还是反向传播过程中，信号都能够不同单元之间直接传递。实验结果也从实际角度证明，当网络架构更接近上述两种条件时，训练过程会变得更加容易。

为了解析跳过连接的作用，我们对各类 $h(\mathbf{x}_l)$ 进行了分析与比较。研究表明，在 [1] 中选用的恒等映射 $h(\mathbf{x}_l) = \mathbf{x}_l$ ，能够实现最快的误差降低效果以及最低的训练损失。

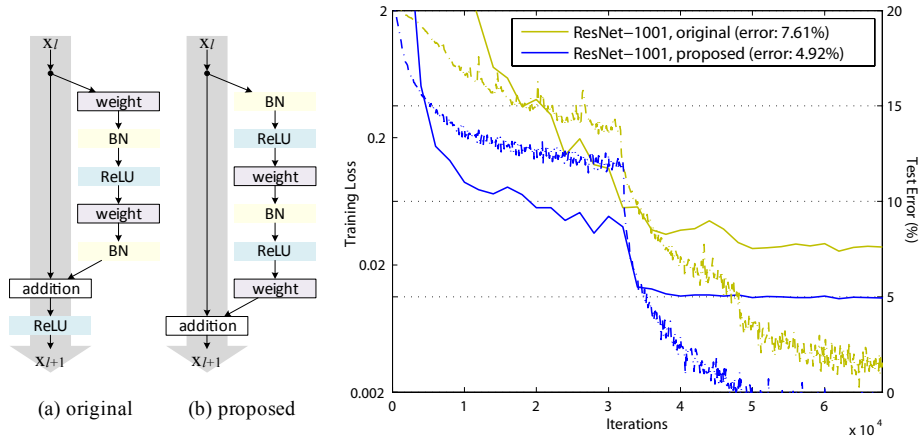


Figure 1. Left: (a) original Residual Unit in [1]; (b) proposed Residual Unit. The grey arrows indicate the easiest paths for the information to propagate, corresponding to the additive term “ $\mathbf{x}_l$ ” in Eqn.(4) (forward propagation) and the additive term “1” in Eqn.(5) (backward propagation). **Right:** training curves on CIFAR-10 of **1001-layer** ResNets. Solid lines denote test error (y-axis on the right), and dashed lines denote training loss (y-axis on the left). The proposed unit makes ResNet-1001 easier to train.

achieves the fastest error reduction and lowest training loss among all variants we investigated, whereas skip connections of scaling, gating [5,6,7], and 1×1 convolutions all lead to higher training loss and error. These experiments suggest that keeping a “clean” information path (indicated by the grey arrows in Fig. 1, 2, and 4) is helpful for easing optimization.

To construct an identity mapping $f(\mathbf{y}_l) = \mathbf{y}_l$, we view the activation functions (ReLU and BN [8]) as “pre-activation” of the weight layers, in contrast to conventional wisdom of “post-activation”. This point of view leads to a new residual unit design, shown in (Fig. 1(b)). Based on this unit, we present competitive results on CIFAR-10/100 with a 1001-layer ResNet, which is much easier to train and generalizes better than the original ResNet in [1]. We further report improved results on ImageNet using a 200-layer ResNet, for which the counterpart of [1] starts to overfit. These results suggest that there is much room to exploit the dimension of *network depth*, a key to the success of modern deep learning.

2 Analysis of Deep Residual Networks

The ResNets developed in [1] are *modularized* architectures that stack building blocks of the same connecting shape. In this paper we call these blocks “Residual

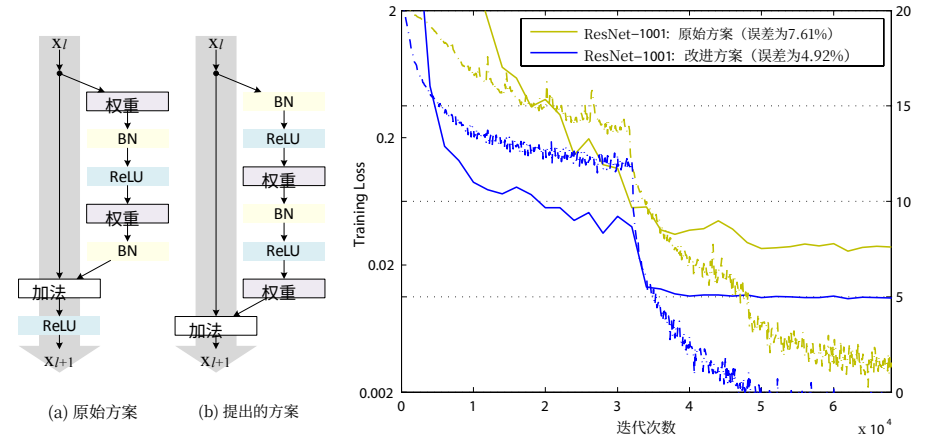


图1. 左侧: (a) [1]中原有的残差单元; (b)所提出的新残差单元。灰色箭头标出了信息传递的最便捷路径，这些路径分别对应方程(4)中的加性项“ $\mathbf{x}_l$ ”（即前向传播过程），以及方程(5)中的加性项“1”（即反向传播过程）。**右侧**展示的是在CIFAR-10数据集上，**1001层**的ResNets对应的训练曲线。实线代表测试误差（位于右侧的y轴），虚线代表训练损失（位于左侧的y轴）。这种新的残差单元使得ResNet-1001更易于训练。

在所有我们测试过的变体中，这种恒等映射方式能够带来最快的误差下降速度和最低的训练损失；而采用缩放机制、门控机制 [5,6,7],以及 1×1 卷积结构的跳过连接，则都会导致更高的训练损失和更大的误差。这些实验表明，保持一条“纯净”的信息传递路径——如图1、图2和图4中的灰色箭头所示——有助于简化优化过程。

为构建恒等映射 $f(\mathbf{y}_l) = \mathbf{y}_l$ ，我们将ReLU激活函数与批归一化 [8]视为权重层的“激活前处理”，这一视角与传统认为的“激活后处理”有所不同。基于这一思路，我们设计出了新的残差单元，如图1(b)所示。利用该单元，我们在CIFAR-10/100数据集上使用了1001层的ResNet进行了实验，结果发现其不仅更易训练，泛化能力也优于原有的ResNet [1]。此外，我们还使用200层的ResNet在ImageNet数据集上取得了更好的性能，因为在那种架构下 [1] 对应的结构容易出现过拟合现象。这些结果表明，充分利用网络深度这一维度仍有很大潜力，而网络深度正是现代深度学习取得成功的关键所在。

2 深度残差网络的分析

在 [1] 中开发的ResNet系列网络是模块化架构，这类架构会堆叠具有相同连接方式的构建单元。在本文中，我们将这些构建单元称为残差单元。

Units”. The original Residual Unit in [1] performs the following computation:

$$\mathbf{y}_l = h(\mathbf{x}_l) + \mathcal{F}(\mathbf{x}_l, \mathcal{W}_l), \quad (1)$$

$$\mathbf{x}_{l+1} = f(\mathbf{y}_l). \quad (2)$$

Here $\mathbf{x}_l$ is the input feature to the l -th Residual Unit. $\mathcal{W}_l = \{\mathcal{W}_{l,k} | 1 \leq k \leq K\}$ is a set of weights (and biases) associated with the l -th Residual Unit, and K is the number of layers in a Residual Unit (K is 2 or 3 in [1]). $\mathcal{F}$ denotes the residual function, *e.g.*, a stack of two 3×3 convolutional layers in [1]. The function f is the operation after element-wise addition, and in [1] f is ReLU. The function h is set as an identity mapping: $h(\mathbf{x}_l) = \mathbf{x}_l$.¹

If f is also an identity mapping: $\mathbf{x}_{l+1} \equiv \mathbf{y}_l$, we can put Eqn.(2) into Eqn.(1) and obtain:

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \mathcal{F}(\mathbf{x}_l, \mathcal{W}_l). \quad (3)$$

Recursively ($\mathbf{x}_{l+2} = \mathbf{x}_{l+1} + \mathcal{F}(\mathbf{x}_{l+1}, \mathcal{W}_{l+1}) = \mathbf{x}_l + \mathcal{F}(\mathbf{x}_l, \mathcal{W}_l) + \mathcal{F}(\mathbf{x}_{l+1}, \mathcal{W}_{l+1})$, etc.) we will have:

$$\mathbf{x}_L = \mathbf{x}_l + \sum_{i=l}^{L-1} \mathcal{F}(\mathbf{x}_i, \mathcal{W}_i), \quad (4)$$

for *any deeper unit* L and *any shallower unit* l . Eqn.(4) exhibits some nice properties. (i) The feature $\mathbf{x}_L$ of any deeper unit L can be represented as the feature $\mathbf{x}_l$ of any shallower unit l plus a residual function in a form of $\sum_{i=l}^{L-1} \mathcal{F}$, indicating that the model is in a *residual* fashion between any units L and l . (ii) The feature $\mathbf{x}_L = \mathbf{x}_0 + \sum_{i=0}^{L-1} \mathcal{F}(\mathbf{x}_i, \mathcal{W}_i)$, of any deep unit L , is the *summation* of the outputs of all preceding residual functions (plus $\mathbf{x}_0$). This is in contrast to a “plain network” where a feature $\mathbf{x}_L$ is a series of matrix-vector *products*, say, $\prod_{i=0}^{L-1} \mathcal{W}_i \mathbf{x}_0$ (ignoring BN and ReLU).

Eqn.(4) also leads to nice backward propagation properties. Denoting the loss function as $\mathcal{E}$, from the chain rule of backpropagation [9] we have:

$$\frac{\partial \mathcal{E}}{\partial \mathbf{x}_l} = \frac{\partial \mathcal{E}}{\partial \mathbf{x}_L} \frac{\partial \mathbf{x}_L}{\partial \mathbf{x}_l} = \frac{\partial \mathcal{E}}{\partial \mathbf{x}_L} \left(1 + \frac{\partial}{\partial \mathbf{x}_l} \sum_{i=l}^{L-1} \mathcal{F}(\mathbf{x}_i, \mathcal{W}_i) \right). \quad (5)$$

Eqn.(5) indicates that the gradient $\frac{\partial \mathcal{E}}{\partial \mathbf{x}_l}$ can be decomposed into two additive terms: a term of $\frac{\partial \mathcal{E}}{\partial \mathbf{x}_L}$ that propagates information directly without concerning any weight layers, and another term of $\frac{\partial \mathcal{E}}{\partial \mathbf{x}_L} \left(\frac{\partial}{\partial \mathbf{x}_l} \sum_{i=l}^{L-1} \mathcal{F} \right)$ that propagates through the weight layers. The additive term of $\frac{\partial \mathcal{E}}{\partial \mathbf{x}_L}$ ensures that information is directly propagated back to *any shallower unit* l . Eqn.(5) also suggests that it

¹ It is noteworthy that there are Residual Units for increasing dimensions and reducing feature map sizes [1] in which h is not identity. In this case the following derivations do not hold strictly. But as there are only a very few such units (two on CIFAR and three on ImageNet, depending on image sizes [1]), we expect that they do not have the exponential impact as we present in Sec. 3. One may also think of our derivations as applied to all Residual Units within the same feature map size.

单元: ”在 [1] 中提出的原始残差单元会执行如下的计算操作。

$$\mathbf{y}_l = h(\mathbf{x}_l) + \mathcal{F}(\mathbf{x}_l, \mathcal{W}_l), \quad (1)$$

$$\mathbf{x}_{l+1} = f(\mathbf{y}_l). \quad (2)$$

此处 $\mathbf{x}_l$ 即为第 l 个残差单元的输入特征。 $\mathcal{W}_l = \{\mathcal{W}_{l,k} | 1 \leq k \leq K\}$ 是一组与第 l 个残差单元相关的权重（以及偏置），而 K 则表示该残差单元的层数（在[1]中为2或3）。 $\mathcal{F}$ 代表残差函数，例如，即在[1]中堆叠的两个 3×3 卷积层。 f 表示逐元素相加之后的运算，在[1] f 中该运算是ReLU函数。至于 h ，则被设定为恒等映射： $h(\mathbf{x}_l) = \mathbf{x}_l$ 。

如果 f 也属于恒等映射，即 $\mathbf{x}_{l+1} \equiv \mathbf{y}_l$ ，那么我们就可以将方程(2)代入方程(1)，进而得到：

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \mathcal{F}(\mathbf{x}_l, \mathcal{W}_l). \quad (3)$$

通过递归方式（ $\mathbf{x}_{l+2} = \mathbf{x}_{l+1} + \mathcal{F}(\mathbf{x}_{l+1}, \mathcal{W}_{l+1}) = \mathbf{x}_l + \mathcal{F}(\mathbf{x}_l, \mathcal{W}_l) + \mathcal{F}(\mathbf{x}_{l+1}, \mathcal{W}_{l+1})$ ），以此类推，最终可以得到：

$$\mathbf{x}_L = \mathbf{x}_l + \sum_{i=l}^{L-1} \mathcal{F}(\mathbf{x}_i, \mathcal{W}_i), \quad (4)$$

对于任意更深的单元 L 以及任意更浅的单元 l 而言，方程(4)展现出一些优良的特性。(i) 任意更深单元 L 的特征 $\mathbf{x}_L$ 可以表示为任意更浅单元 l 的特征 $\mathbf{x}_l$ 再加上形式为 $\sum_{i=l}^{L-1} \mathcal{F}$ 的残差函数，这表明在任意单元 L 与 l 之间，模型采用的是残差结构。(ii) 任意深层单元 L 的特征 $\mathbf{x}_L = \mathbf{x}_0 + \sum_{i=0}^{L-1} \mathcal{F}$ （即 $\mathbf{x}_i, \mathcal{W}_i$ ）实际上等于所有前置残差函数的输出之和（再加上 $\mathbf{x}_0$ ）。这与“普通网络”有所不同，在普通网络中，特征 $\mathbf{x}_L$ 是一系列矩阵-向量乘积的结果，例如 $\prod_{i=0}^{L-1} \mathcal{W}_i \mathbf{x}_0$ （此处不考虑批归一化与ReLU的作用）。

方程(4)还能带来良好的反向传播特性。将损失函数记为 $\mathcal{E}$ ，根据反向传播的链式法则[9]，我们可以得到：

$$\frac{\partial \mathcal{E}}{\partial \mathbf{x}_l} = \frac{\partial \mathcal{E}}{\partial \mathbf{x}_L} \frac{\partial \mathbf{x}_L}{\partial \mathbf{x}_l} = \frac{\partial \mathcal{E}}{\partial \mathbf{x}_L} \left(1 + \frac{\partial}{\partial \mathbf{x}_l} \sum_{i=l}^{L-1} \mathcal{F}(\mathbf{x}_i, \mathcal{W}_i) \right). \quad (5)$$

方程(5)表明，损失函数 $\mathcal{E}$ 的梯度 $\frac{\partial \mathcal{E}}{\partial \mathbf{x}_l}$ 可以分解为两个相加的项：一项是 $\frac{\partial \mathcal{E}}{\partial \mathbf{x}_L}$ ，它无需经过任何权重层即可直接传递信息；另一项则是 $\frac{\partial \mathcal{E}}{\partial \mathbf{x}_L} \left(\frac{\partial}{\partial \mathbf{x}_l} \sum_{i=l}^{L-1} \mathcal{F} \right)$ ，它需要通过权重层来传递信息。 $\frac{\partial \mathcal{E}}{\partial \mathbf{x}_L}$ 对应的那个相加项能够确保信息直接传回任意较浅层的单元 l 。方程(5)还进一步指出，对于一个小批量数据而言，梯度 $\frac{\partial \mathcal{E}}{\partial \mathbf{x}_l}$ 不太可能被完全抵消，因为通常情况下 $\frac{\partial \mathcal{E}}{\partial \mathbf{x}_L}$ 并不可能对小批量数据中的所有样本都恒等于-1。这意味着，即便权重数值极小，层的梯度也不会消失。

¹ 值得注意的是，存在用于提升维度并缩小特征图尺寸的[1]残差单元，而其中的 h 并非恒等映射函数。在这种情况下，后续的推导并不完全成立。但由于这类单元的数量极为有限（在CIFAR数据集中为两个，在ImageNet数据集中根据图像尺寸不同为三个[1]），我们预计它们不会像我们在第3节中所阐述的那样产生指数级的影响。人们也可以将这些推导视为适用于同一特征图尺寸下的所有残差单元。

is unlikely for the gradient $\frac{\partial \mathcal{E}}{\partial \mathbf{x}_l}$ to be canceled out for a mini-batch, because in general the term $\frac{\partial}{\partial \mathbf{x}_l} \sum_{i=l}^{L-1} \mathcal{F}$ cannot be always -1 for all samples in a mini-batch. This implies that the gradient of a layer does not vanish even when the weights are arbitrarily small.

Discussions

Eqn.(4) and Eqn.(5) suggest that the signal can be directly propagated from any unit to another, both forward and backward. The foundation of Eqn.(4) is two identity mappings: (i) the identity skip connection $h(\mathbf{x}_l) = \mathbf{x}_l$, and (ii) the condition that f is an identity mapping.

These directly propagated information flows are represented by the grey arrows in Fig. 1, 2, and 4. And the above two conditions are true when these grey arrows cover no operations (except addition) and thus are “clean”. In the following two sections we separately investigate the impacts of the two conditions.

3 On the Importance of Identity Skip Connections

Let’s consider a simple modification, $h(\mathbf{x}_l) = \lambda_l \mathbf{x}_l$, to break the identity shortcut:

$$\mathbf{x}_{l+1} = \lambda_l \mathbf{x}_l + \mathcal{F}(\mathbf{x}_l, \mathcal{W}_l), \quad (6)$$

where λ_l is a modulating scalar (for simplicity we still assume f is identity). Recursively applying this formulation we obtain an equation similar to Eqn. (4): $\mathbf{x}_L = (\prod_{i=l}^{L-1} \lambda_i) \mathbf{x}_l + \sum_{i=l}^{L-1} (\prod_{j=i+1}^{L-1} \lambda_j) \mathcal{F}(\mathbf{x}_i, \mathcal{W}_i)$, or simply:

$$\mathbf{x}_L = \left(\prod_{i=l}^{L-1} \lambda_i \right) \mathbf{x}_l + \sum_{i=l}^{L-1} \hat{\mathcal{F}}(\mathbf{x}_i, \mathcal{W}_i), \quad (7)$$

where the notation $\hat{\mathcal{F}}$ absorbs the scalars into the residual functions. Similar to Eqn.(5), we have backpropagation of the following form:

$$\frac{\partial \mathcal{E}}{\partial \mathbf{x}_l} = \frac{\partial \mathcal{E}}{\partial \mathbf{x}_L} \left(\left(\prod_{i=l}^{L-1} \lambda_i \right) + \frac{\partial}{\partial \mathbf{x}_l} \sum_{i=l}^{L-1} \hat{\mathcal{F}}(\mathbf{x}_i, \mathcal{W}_i) \right). \quad (8)$$

Unlike Eqn.(5), in Eqn.(8) the first additive term is modulated by a factor $\prod_{i=l}^{L-1} \lambda_i$. For an extremely deep network (L is large), if $\lambda_i > 1$ for all i , this factor can be exponentially large; if $\lambda_i < 1$ for all i , this factor can be exponentially small and vanish, which blocks the backpropagated signal from the shortcut and forces it to flow through the weight layers. This results in optimization difficulties as we show by experiments.

In the above analysis, the original identity skip connection in Eqn.(3) is replaced with a simple scaling $h(\mathbf{x}_l) = \lambda_l \mathbf{x}_l$. If the skip connection $h(\mathbf{x}_l)$ represents more complicated transforms (such as gating and 1×1 convolutions), in Eqn.(8) the first term becomes $\prod_{i=l}^{L-1} h'_i$ where h' is the derivative of h . This product may also impede information propagation and hamper the training procedure as witnessed in the following experiments.

这是因为对于一个小批量数据来说，梯度 $\frac{\partial \mathcal{E}}{\partial \mathbf{x}_l}$ 很难被完全抵消，毕竟 $\frac{\partial}{\partial \mathbf{x}_l} \sum_{i=l}^{L-1} \mathcal{F}$ 通常不可能在所有样本上始终等于-1。由此可知，即使权重数值非常小，层的梯度依然不会消失。

讨论

方程(4)与方程(5)表明，信号能够在前向和反向传播过程中直接从任意一个单元传递到另一个单元。方程(4)的成立基于两种恒等映射：(i)恒等跳连接 $h(\mathbf{x}_l) = \mathbf{x}_l$ ，以及(ii) f 为恒等映射这一条件。

图1、图2和图4中的灰色箭头分别代表了这些直接传递的信息流。当这些灰色箭头未经过任何运算（加法除外）时，上述两个条件即成立，此时信息流可视为“纯净”的。在接下来的两个章节中，我们将分别探讨这两个条件所产生的影响。

3 恒等跳连接的重要性

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让我们考虑一种简单的修改方式，即对 $h(\mathbf{x}_l) = \lambda_l \mathbf{x}_l$ 进行修改，以此打破原有的恒等捷径。

$$\mathbf{x}_{l+1} = \lambda_l \mathbf{x}_l + \mathcal{F}(\mathbf{x}_l, \mathcal{W}_l), \quad (6)$$

其中 λ_l 是一个调制标量（为简化分析，我们仍假设 f 为恒等映射）。通过递归应用这一公式，我们可以得到一个与方程(4)类似的表达式： $\mathbf{x}_L = (\prod_{i=l}^{L-1} \lambda_i) \mathbf{x}_l + \sum_{i=l}^{L-1} (\prod_{j=i+1}^{L-1} \lambda_j) \mathcal{F}(\mathbf{x}_i, \mathcal{W}_i)$ ，或者更简而言之就是如此。

$$\mathbf{x}_L = \left(\prod_{i=l}^{L-1} \lambda_i \right) \mathbf{x}_l + \sum_{i=l}^{L-1} \hat{\mathcal{F}}(\mathbf{x}_i, \mathcal{W}_i), \quad (7)$$

其中符号 $\mathcal{F}$ 用于将标量值整合到残差函数中。与方程(5)类似，反向传播的过程也呈现为如下形式：

$$\frac{\partial \mathcal{E}}{\partial \mathbf{x}_l} = \frac{\partial \mathcal{E}}{\partial \mathbf{x}_L} \left(\left(\prod_{i=l}^{L-1} \lambda_i \right) + \frac{\partial}{\partial \mathbf{x}_l} \sum_{i=l}^{L-1} \hat{\mathcal{F}}(\mathbf{x}_i, \mathcal{W}_i) \right). \quad (8)$$

不同于方程(5)，在方程(8)中第一个加法项会受到一个系数 $\prod_{i=l}^{L-1} \lambda_i$ 的调节。对于极其深度的网络（即 L 数值很大），若对所有 i 都有 $\lambda_i > 1$ 成立，则该系数可能达到指数级大；而若对所有 i 都有 $\lambda_i < 1$ 成立，该系数则可能变得指数级小甚至趋近于零，这样一来，从捷径路径传来的信号就会被阻断，只能通过权重层传递。正如我们通过实验所展示的，这种情况会导致优化过程出现困难。

在上述分析中，方程(3)中所使用的原始恒等跳连接被替换为一种简单的缩放机制 $h(\mathbf{x}_l) = \lambda_l \mathbf{x}_l$ 。如果该跳连接 $h(\mathbf{x}_l)$ 涉及更复杂的变换（如门控机制以及 1×1 卷积操作），那么在方程(8)中第一项就会变为 $\prod_{i=l}^{L-1} h'_i$ ，此处 h' 表示 h 的导数。正如后续实验所显示的那样，这种乘积结构可能会阻碍信息传播，进而影响训练进程。

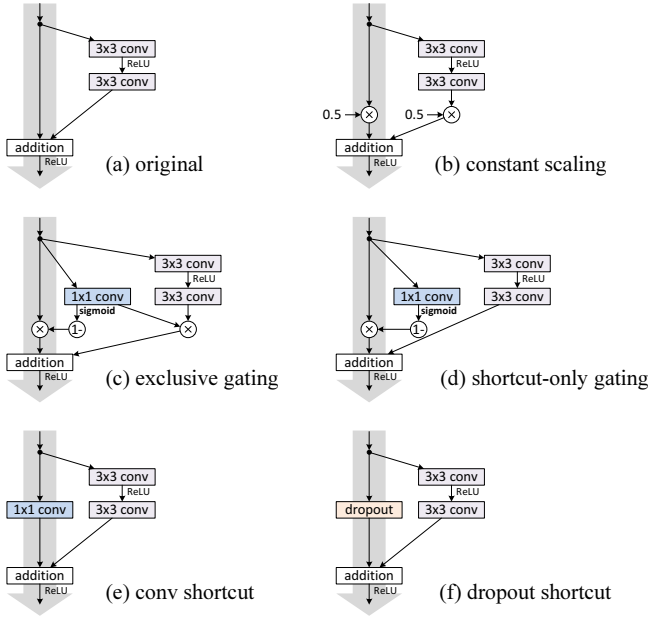


Figure 2. Various types of shortcut connections used in Table 1. The grey arrows indicate the easiest paths for the information to propagate. The shortcut connections in (b-f) are impeded by different components. For simplifying illustrations we do not display the BN layers, which are adopted right after the weight layers for all units here.

3.1 Experiments on Skip Connections

We experiment with the 110-layer ResNet as presented in [1] on CIFAR-10 [10]. This extremely deep ResNet-110 has 54 two-layer Residual Units (consisting of 3×3 convolutional layers) and is challenging for optimization. Our implementation details (see appendix) are the same as [1]. Throughout this paper we report the median accuracy of **5 runs** for each architecture on CIFAR, reducing the impacts of random variations.

Though our above analysis is driven by identity f , the experiments in this section are all based on $f = \text{ReLU}$ as in [1]; we address identity f in the next section. Our baseline ResNet-110 has 6.61% error on the test set. The comparisons of other variants (Fig. 2 and Table 1) are summarized as follows:

Constant scaling. We set $\lambda = 0.5$ for all shortcuts (Fig. 2(b)). We further study two cases of scaling $\mathcal{F}$: (i) $\mathcal{F}$ is not scaled; or (ii) $\mathcal{F}$ is scaled by a constant scalar of $1 - \lambda = 0.5$, which is similar to the highway gating [6,7] but with frozen gates. The former case does not converge well; the latter is able to converge, but the test error (Table 1, 12.35%) is substantially higher than the original ResNet-110. Fig 3(a) shows that the training error is higher than that of the original ResNet-110, suggesting that the optimization has difficulties when the shortcut signal is scaled down.

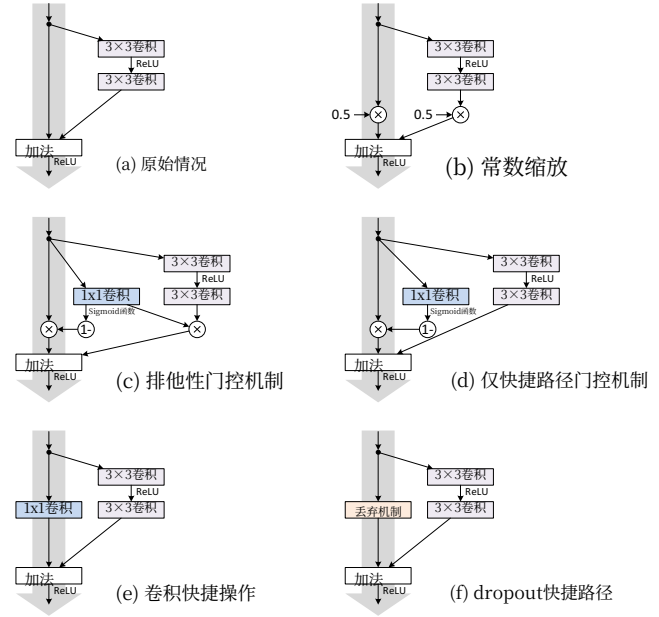


图2. 表1中展示了多种类型的快捷连接方式。灰色箭头标示出信息传递的最便捷路径，而(b-f)中的快捷连接则会受到不同组件的干扰。为简化示意图，此处未显示所有单元对应的权重层之后的批归一化层。

3.1 跳过连接的实验

我们采用了 [1] 中介绍的、具有110层的ResNet网络，在CIFAR-10 [10]数据集上开展了相关实验。这款深度极高的ResNet-110网络包含54个由 3×3 卷积层构成的双层残差单元，因此其优化过程颇具挑战性。我们的实现细节（详见附录）与 [1]中的描述一致。在本文中，我们对每种架构在CIFAR数据集上的**5次测试**结果给出了中位数准确率，以此降低随机因素带来的影响。

尽管上述分析是以恒等映射函数 f 为理论基础，但本节中的所有实验均采用 [1]中所述的 $f = \text{ReLU}$ 作为损失函数 E ，关于恒等映射函数 f 的内容我们将在下一节探讨。我们的基准方案ResNet-110在测试集上的误差率为6.61%，其他变体的对比情况（见图2与表1）总结如下：

常数缩放机制。我们对所有快捷路径均设定为 $\lambda = 0.5$ （见图2(b)）。随后我们研究了两种 $\mathcal{F}$ 缩放方式：(i) $\mathcal{F}$ 不进行缩放；或(ii) $\mathcal{F}$ 通过常数标量 $1 - \lambda = 0.5$ 进行缩放，这种方式与Highway门控机制 [6,7] 类似，只不过其门控单元处于冻结状态。前者无法良好收敛，而后者虽能收敛，但其测试误差（见表1，为12.35%）仍远高于原始的ResNet-110网络。图3(a)显示，该网络的训练误差也高于原始ResNet-110，这表明当快捷路径信号被缩小后，优化过程会面临较大困难。

Table 1. Classification error on the CIFAR-10 test set using ResNet-110 [1], with different types of shortcut connections applied to all Residual Units. We report “fail” when the test error is higher than 20%.

case	Fig.	on shortcut	on $\mathcal{F}$	error (%)	remark
original [1]	Fig. 2(a)	1	1	6.61	
constant scaling	Fig. 2(b)	0	1	fail	This is a plain net
		0.5	1	fail	
		0.5	0.5	12.35	frozen gating
exclusive gating	Fig. 2(c)	$1 - g(\mathbf{x})$	$g(\mathbf{x})$	fail	init $b_g=0$ to -5
		$1 - g(\mathbf{x})$	$g(\mathbf{x})$	8.70	init $b_g=-6$
		$1 - g(\mathbf{x})$	$g(\mathbf{x})$	9.81	init $b_g=-7$
shortcut-only gating	Fig. 2(d)	$1 - g(\mathbf{x})$	1	12.86	init $b_g=0$
		$1 - g(\mathbf{x})$	1	6.91	init $b_g=-6$
1×1 conv shortcut	Fig. 2(e)	1×1 conv	1	12.22	
dropout shortcut	Fig. 2(f)	dropout 0.5	1	fail	

Exclusive gating. Following the Highway Networks [6,7] that adopt a gating mechanism [5], we consider a gating function $g(\mathbf{x}) = \sigma(W_g \mathbf{x} + b_g)$ where a transform is represented by weights W_g and biases b_g followed by the sigmoid function $\sigma(x) = \frac{1}{1+e^{-x}}$. In a convolutional network $g(\mathbf{x})$ is realized by a 1×1 convolutional layer. The gating function modulates the signal by element-wise multiplication.

We investigate the “exclusive” gates as used in [6,7] — the $\mathcal{F}$ path is scaled by $g(\mathbf{x})$ and the shortcut path is scaled by $1 - g(\mathbf{x})$. See Fig 2(c). We find that the initialization of the biases b_g is critical for training gated models, and following the guidelines² in [6,7], we conduct hyper-parameter search on the initial value of b_g in the range of 0 to -10 with a decrement step of -1 on the training set by cross-validation. The best value (-6 here) is then used for training on the training set, leading to a test result of 8.70% (Table 1), which still lags far behind the ResNet-110 baseline. Fig 3(b) shows the training curves. Table 1 also reports the results of using other initialized values, noting that the exclusive gating network does not converge to a good solution when b_g is not appropriately initialized.

The impact of the exclusive gating mechanism is two-fold. When $1 - g(\mathbf{x})$ approaches 1, the gated shortcut connections are closer to identity which helps information propagation; but in this case $g(\mathbf{x})$ approaches 0 and suppresses the function $\mathcal{F}$. To isolate the effects of the gating functions on the shortcut path alone, we investigate a non-exclusive gating mechanism in the next.

Shortcut-only gating. In this case the function $\mathcal{F}$ is not scaled; only the shortcut path is gated by $1 - g(\mathbf{x})$. See Fig 2(d). The initialized value of b_g is still essential in this case. When the initialized b_g is 0 (so initially the expectation of $1 - g(\mathbf{x})$ is 0.5), the network converges to a poor result of 12.86% (Table 1). This is also caused by higher training error (Fig 3(c)).

² See also: people.idsia.ch/~rupesh/very_deep_learning/ by [6,7].

表1. 展示使用ResNet-110 [1], 并在所有残差单元中应用不同类型快捷连接时的CIFAR-10测试集分类误差情况。若测试误差超过20%，则记录为“失败”。

case	Fig.	关于快捷连接	on $\mathcal{F}$	误差百分比	备注
原始内容 [1]	图2(a)	1	1	6.61	
常数 缩放机制	图2(b)	0	1	fail	这是一个简单的神经网络。
		0.5	1	fail	
		0.5	0.5	12.35	冻结的门控机制
唯一的 门控机制	图2(c)	$1 - g(\mathbf{x})$	$g(\mathbf{x})$	fail	初始化参数 $b_g=0$ 到 -5
		$1 - g(\mathbf{x})$	$g(\mathbf{x})$	8.70	初始化参数 $b_g=-6$
		$1 - g(\mathbf{x})$	$g(\mathbf{x})$	9.81	初始化参数 $b_g=-7$
仅快捷方式 门控机制	图2(d)	$1 - g(\mathbf{x})$	1	12.86	初始化参数 $b_g=0$
		$1 - g(\mathbf{x})$	1	6.91	初始化参数 $b_g=-6$
1×1 卷积快捷操作	图2(e)	1×1 conv	1	12.22	
dropout快捷路径	图2(f)	dropout系数为0.5	1	fail	

排他性门控机制。在那些采用门控机制 [5], 的Highway Networks [6,7] 基础上, 我们引入了一种门控函数 $g(\mathbf{x}) = \sigma(W_g \mathbf{x} + b_g)$, 其中变换操作由权重 W_g 和偏置 b_g 共同实现, 之后再经过Sigmoid函数 $\sigma(x) = \frac{1}{1+e^{-x}}$ 处理。在卷积网络 g 中, $(\mathbf{x})$ 是通过某个 1×1 卷积层来实现的。该门控函数通过逐元素乘法的方式对信号进行调节。

我们研究了 [6,7] 中使用的“排他性”门控机制—— $\mathcal{F}$ 路径会通过 $g(\mathbf{x})$ 进行缩放, 而快捷路径则通过 $1 - g(\mathbf{x})$ 进行缩放, 具体可见图2(c)。研究表明, 偏置项 b_g 的初始化对训练带有门控机制的模型至关重要。依照 [6,7], 中的指导原则², 我们在交叉验证的训练集上对 b_g 的初始值进行了超参数搜索, 搜索范围为0到-10, 步长为-1。最终选出的最佳值 (此处为 -6) 被用于训练集训练, 所得测试结果为 8.70% (见表1), 这一数值仍远低于ResNet-110基准模型的表现。图3(b)展示了相应的训练曲线。表1还列出了使用其他初始化值时的测试结果, 从中可以看出, 若 b_g 的初始化不当, 排他性门控网络就无法收敛到理想的解决方案。

排他性门控机制的作用体现在两个方面。当 $1 - g(\mathbf{x})$ 趋近于1时, 经过门控控制的快捷路径更接近恒等映射函数, 从而有助于信息的传播; 但此时 $g(\mathbf{x})$ 则趋近于0, 进而抑制 $\mathcal{F}$ 所对应的函数作用。为单独研究门控机制对快捷路径的影响, 我们将在后续内容中探讨非排他性门控机制。

仅快捷路径门控机制。在这种情形下, 函数 $\mathcal{F}$ 不会被缩放处理, 仅有快捷路径会通过 $1 - g(\mathbf{x})$ 进行门控控制。详见图2(d)。在此情况下, b_g 的初始化值依然十分重要。当 b_g 的初始值为0时 (这样一来 $1 - g(\mathbf{x})$ 的初始期望值即为0.5), 网络会收敛到 12.86% 这一较差的结果 (见表1), 而这同样是由于训练误差较高所致 (见图3(c))。

² 另请参阅: people.idsia.ch/~rupesh/very_deep_learning/ 相关资料。

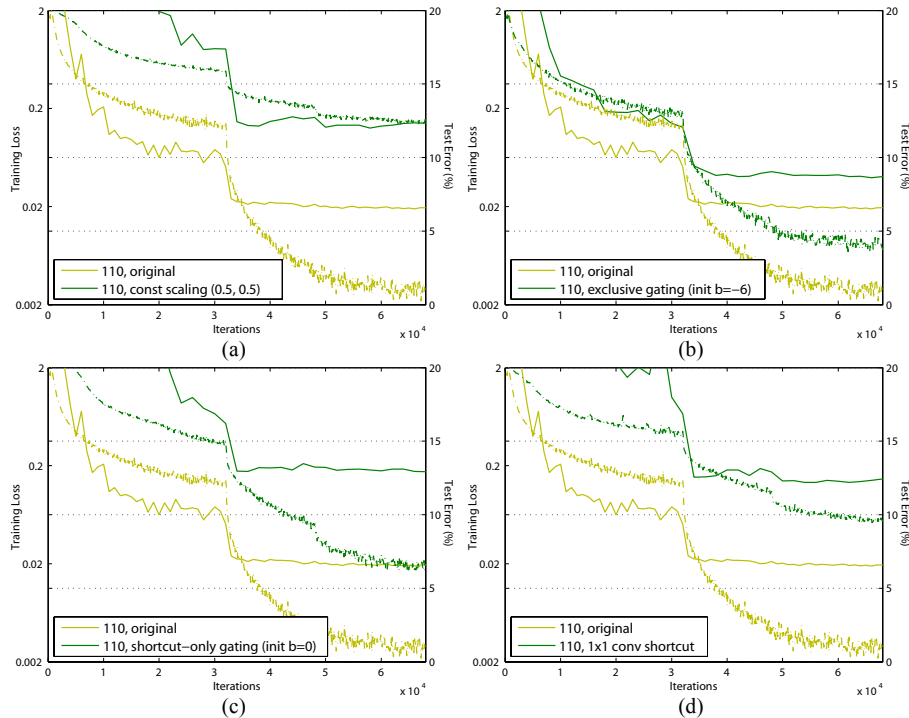


Figure 3. Training curves on CIFAR-10 of various shortcuts. Solid lines denote test error (y-axis on the right), and dashed lines denote training loss (y-axis on the left).

When the initialized b_g is very negatively biased (*e.g.*, -6), the value of $1 - g(\mathbf{x})$ is closer to 1 and the shortcut connection is nearly an identity mapping. Therefore, the result (6.91%, Table 1) is much closer to the ResNet-110 baseline.

1×1 convolutional shortcut. Next we experiment with 1×1 convolutional shortcut connections that replace the identity. This option has been investigated in [1] (known as option C) on a 34-layer ResNet (16 Residual Units) and shows good results, suggesting that 1×1 shortcut connections could be useful. But we find that this is not the case when there are many Residual Units. The 110-layer ResNet has a poorer result (12.22%, Table 1) when using 1×1 convolutional shortcuts. Again, the training error becomes higher (Fig 3(d)). When stacking so many Residual Units (54 for ResNet-110), even the shortest path may still impede signal propagation. We witnessed similar phenomena on ImageNet with ResNet-101 when using 1×1 convolutional shortcuts.

Dropout shortcut. Last we experiment with dropout [11] (at a ratio of 0.5) which we adopt on the output of the identity shortcut (Fig. 2(f)). The network fails to converge to a good solution. Dropout statistically imposes a scale of λ with an expectation of 0.5 on the shortcut, and similar to constant scaling by 0.5, it impedes signal propagation.

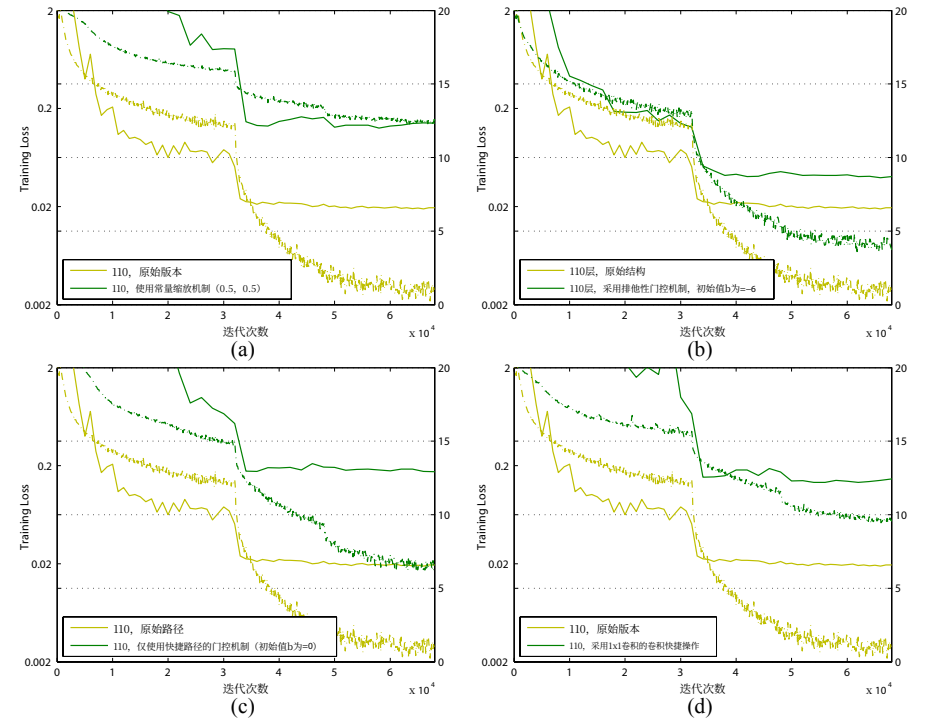


图3. 不同快捷路径在CIFAR-10数据集上的训练曲线。实线表示测试误差（位于右侧纵轴），虚线则表示训练损失（位于左侧纵轴）。

当 b_g 的初始化值存在极大的负偏置时（例如， -6 ）， $1 - g(\mathbf{x})$ 的值会更接近1，此时快捷路径连接几乎相当于恒等映射。因此，所得结果（6.91%，见表1）会更为接近ResNet-110基准方案的结果。

1×1 卷积型快捷路径。 接下来我们尝试使用 1×1 卷积型快捷连接来替代恒等映射函数。这种方案曾在 [1] 中（被称为选项C）于具有34层结构且包含16个残差单元的ResNet网络上被研究过，结果表现良好，这表明 1×1 这类快捷连接可能具有实用价值。然而我们发现，当残差单元数量较多时情况并非如此。在采用 1×1 卷积型快捷路径的110层ResNet网络中，其测试误差较高（为12.22%，详见表1），且训练误差也会随之上升（见图3(d)）。即便在堆叠了众多残差单元（ResNet-110共有54个）的情况下，最短的路径仍可能阻碍信号的传播。在使用 1×1 卷积型快捷路径的ResNet-101网络于ImageNet数据集上时，我们也观察到了类似的现象。

dropout快捷路径。 在上一次的实验中，我们采用了丢弃机制 [11]（激活率为0.5），并将其应用于恒等映射快捷路径的输出上（图2(f)）。但该网络未能收敛到理想的解。从统计角度来看，丢弃机制会为这类快捷路径施加规模为 λ 、期望值为0.5的约束，这与0.5的常数缩放机制类似，都会阻碍信号的信息传播。

Table 2. Classification error (%) on the CIFAR-10 test set using different activation functions.

case	Fig.	ResNet-110	ResNet-164
original Residual Unit [1]	Fig. 4(a)	6.61	5.93
BN after addition	Fig. 4(b)	8.17	6.50
ReLU before addition	Fig. 4(c)	7.84	6.14
ReLU-only pre-activation	Fig. 4(d)	6.71	5.91
full pre-activation	Fig. 4(e)	6.37	5.46

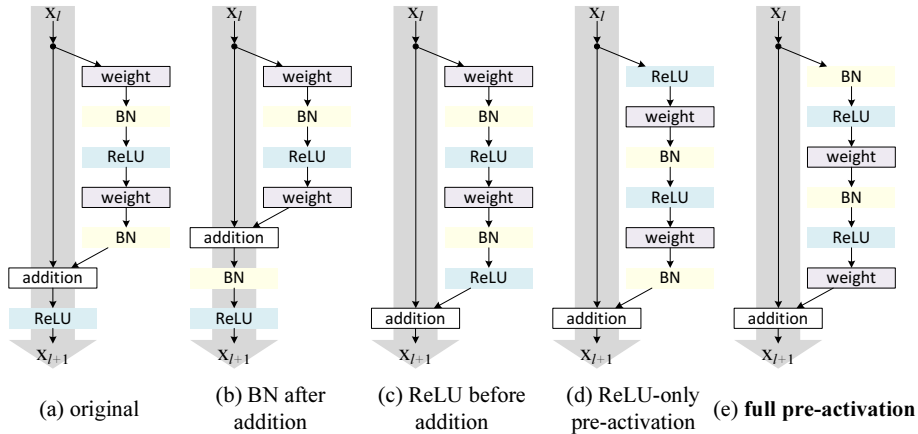


Figure 4. Various usages of activation in Table 2. All these units consist of the same components — only the orders are different.

3.2 Discussions

As indicated by the grey arrows in Fig. 2, the shortcut connections are the most direct paths for the information to propagate. *Multiplicative* manipulations (scaling, gating, 1×1 convolutions, and dropout) on the shortcuts can hamper information propagation and lead to optimization problems.

It is noteworthy that the gating and 1×1 convolutional shortcuts introduce more parameters, and should have stronger *representational* abilities than identity shortcuts. In fact, the shortcut-only gating and 1×1 convolution cover the solution space of identity shortcuts (*i.e.*, they could be optimized as identity shortcuts). However, their training error is higher than that of identity shortcuts, indicating that the degradation of these models is caused by optimization issues, instead of representational abilities.

4 On the Usage of Activation Functions

Experiments in the above section support the analysis in Eqn.(5) and Eqn.(8), both being derived under the assumption that the after-addition activation f

表2. 使用不同激活函数在CIFAR-10测试集上的分类误差百分比。

case	Fig.	ResNet-110网络	ResNet-164架构
传统的残差单元 [1]	图4(a)	6.61	5.93
加法后归一化	图4(b)	8.17	6.50
加法操作之前的ReLU函数	图4(c)	7.84	6.14
仅应用ReLU激活函数的激活前处理	图4(d)	6.71	5.91
完全预激活	图4(e)	6.37	5.46

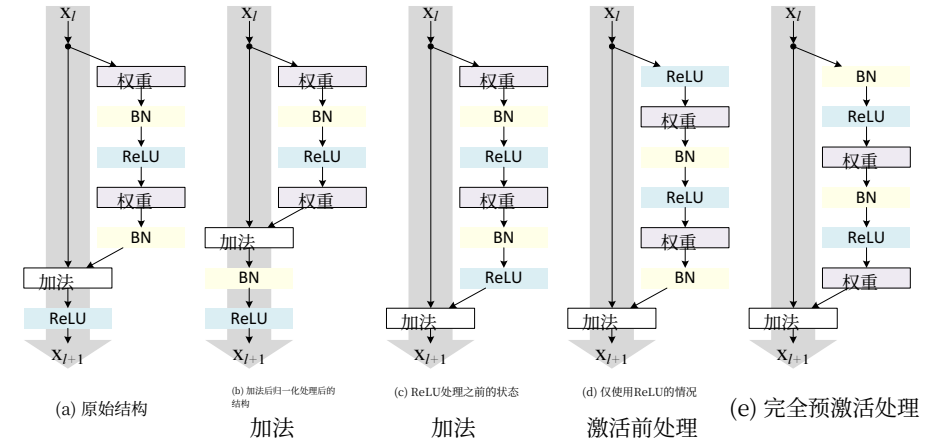


图4. 表2展示了激活函数的多种应用方式。所有这些单元均由相同的组件构成——仅有结构顺序有所不同。

3.2 讨论

如图2中的灰色箭头所示，快捷路径连接是信号传播最直接的路径。乘法操作（包括缩放机制、门控机制、 1×1 卷积以及丢弃机制）作用于这些快捷路径时，都可能干扰信号的信息传播，进而引发优化问题。

值得注意的是，门控机制以及 1×1 卷积型快捷路径会引入更多的网络参数，因此相比恒等映射型快捷路径，它们应具备更强的表征能力。实际上，仅采用快捷路径门控机制的架构以及 1×1 卷积结构能够覆盖恒等映射型快捷路径的解空间（也就是说，它们完全可以被优化为恒等映射型快捷路径）。然而，这类模型的训练误差却高于恒等映射型快捷路径，这表明这些模型的性能下降是由优化问题而非表征能力不足所导致的。

4 激活函数的应用

上一节中的实验为方程(5)和方程(8)中的分析提供了支撑，而这些分析都是在加法后的激活函数 f 为恒等映射函数的假设下推导出来的。

is the identity mapping. But in the above experiments f is ReLU as designed in [1], so Eqn.(5) and (8) are approximate in the above experiments. Next we investigate the impact of f .

We want to make f an identity mapping, which is done by re-arranging the activation functions (ReLU and/or BN). The original Residual Unit in [1] has a shape in Fig. 4(a) — BN is used after each weight layer, and ReLU is adopted after BN except that the last ReLU in a Residual Unit is after element-wise addition ($f = \text{ReLU}$). Fig. 4(b-e) show the alternatives we investigated, explained as following.

4.1 Experiments on Activation

In this section we experiment with ResNet-110 and a 164-layer *Bottleneck* [1] architecture (denoted as ResNet-164). A bottleneck Residual Unit consist of a 1×1 layer for reducing dimension, a 3×3 layer, and a 1×1 layer for restoring dimension. As designed in [1], its computational complexity is similar to the two- 3×3 Residual Unit. More details are in the appendix. The baseline ResNet-164 has a competitive result of 5.93% on CIFAR-10 (Table 2).

BN after addition. Before turning f into an identity mapping, we go the opposite way by adopting BN after addition (Fig. 4(b)). In this case f involves BN and ReLU. The results become considerably worse than the baseline (Table 2). Unlike the original design, now the BN layer alters the signal that passes through the shortcut and impedes information propagation, as reflected by the difficulties on reducing training loss at the beginning of training (Fig. 6 left).

ReLU before addition. A naïve choice of making f into an identity mapping is to move the ReLU before addition (Fig. 4(c)). However, this leads to a *non-negative* output from the transform $\mathcal{F}$, while intuitively a “residual” function should take values in $(-\infty, +\infty)$. As a result, the forward propagated signal is monotonically increasing. This may impact the representational ability, and the result is worse (7.84%, Table 2) than the baseline. We expect to have a residual function taking values in $(-\infty, +\infty)$. This condition is satisfied by other Residual Units including the following ones.

Post-activation or pre-activation? In the original design (Eqn.(1) and Eqn.(2)), the activation $\mathbf{x}_{l+1} = f(\mathbf{y}_l)$ affects *both paths* in the *next* Residual Unit: $\mathbf{y}_{l+1} = f(\mathbf{y}_l) + \mathcal{F}(f(\mathbf{y}_l), \mathcal{W}_{l+1})$. Next we develop an *asymmetric* form where an activation $\hat{f}$ only affects the $\mathcal{F}$ path: $\mathbf{y}_{l+1} = \mathbf{y}_l + \mathcal{F}(\hat{f}(\mathbf{y}_l), \mathcal{W}_{l+1})$, for any l (Fig. 5 (a) to (b)). By renaming the notations, we have the following form:

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \mathcal{F}(\hat{f}(\mathbf{x}_l), \mathcal{W}_l), \quad (9)$$

It is easy to see that Eqn.(9) is similar to Eqn.(4), and can enable a backward formulation similar to Eqn.(5). For this new Residual Unit as in Eqn.(9), the new after-addition activation becomes an identity mapping. This design means that if a new after-addition activation $\hat{f}$ is asymmetrically adopted, it is equivalent to recasting $\hat{f}$ as the *pre-activation* of the next Residual Unit. This is illustrated in Fig. 5.

在上述实验中，加法后的激活函数 f 实际上采用的是 [1] 中所设计的ReLU函数，因此方程(5)和方程(8)在这些实验中的结果仅为近似值。接下来我们将研究 f 所带来的影响。

我们希望实现 f 恒等映射，这一目标可通过重新安排激活函数（ReLU和/或批归一化）来达成。[1]中原有的残差单元结构如图4(a)所示——每个权重层之后都会应用批归一化，而在批归一化之后则使用ReLU激活函数，不过在残差单元中的最后一个ReLU是在各元素进行加法运算之后再应用的（即 $f = \text{ReLU}$ ）。图4(b-e)展示了我们研究过的其他替代方案，具体说明如下。

4.1 激活函数的实验

在本节中，我们以ResNet-110以及具有164层的瓶颈型 [1]架构（记为ResNet-164）作为实验对象。这种瓶颈型残差单元包含一个用于降维的 1×1 层、一个 3×3 层，以及一个用于恢复维度的 1×1 层。根据 [1] 中的设计，它的计算复杂度与传统的双 3×3 残差单元相当。更多细节可见附录。作为基准方案的ResNet-164在CIFAR-10数据集上的测试结果为5.93%，表现十分出色（见表2）。

加法后归一化。在将 f 转化为恒等映射之前，我们采取了相反的做法，即采用加法后归一化方式（图4(b)）。这种情况下 f 会涉及批归一化与ReLU激活函数，其结果相比基准方案要差很多（表2）。与原始设计不同，此时的批归一化层会改变通过捷径传递的信号，从而阻碍信息传播，这一点从训练初期难以降低训练损失的情况中就可以看出来（图6左侧）。

加法前的ReLU。将 f 设为恒等映射的一种简单做法是将ReLU放在加法之前（图4(c)）。然而这样一来，变换 $\mathcal{F}$ 的输出就会变为非负数，而按照直觉，“残差”函数的值应当位于 $(-\infty, +\infty)$ 区间内。因此，正向传播的信号会呈单调递增趋势，这可能会影响模型的表征能力，最终得到的结果（7.84%，表2）甚至比基准方案还要差。我们期望残差函数的值能够处于 $(-\infty, +\infty)$ 范围内，其他残差单元也满足这一条件，其中包括以下几种。

应当进行激活后处理还是激活前处理？在原始设计中（即方程(1)和方程(2)），激活函数 $\mathbf{x}_{l+1} = f(\mathbf{y}_l)$ 会同时影响下一个残差单元中的两条路径，其表达式为： $\mathbf{y}_{l+1} = f(\mathbf{y}_l) + \mathcal{F}(f(\mathbf{y}_l), \mathcal{W}_{l+1})$ 。接下来我们提出了一种非对称的表达形式 $\hat{f}$ ，在这种形式下，激活函数 f 仅会影响其中一条路径，其表达式为： $\mathbf{y}_{l+1} = \mathbf{y}_l + \mathcal{F}(\hat{f}(\mathbf{y}_l), \mathcal{W}_{l+1})$ ，适用于任意的 l （见图5的(a)至(b)）。通过重新命名相关符号，即可得到如下形式。

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \mathcal{F}(\hat{f}(\mathbf{x}_l), \mathcal{W}_l), \quad (9)$$

不难看出，方程(9)与方程(4)十分相似，因此能够推导出类似方程(5)的反向公式。对于方程(9)中所描述的这种新型残差单元，其加法后的激活函数会变为恒等映射函数。这一设计意味着，如果采用非对称形式的加法后激活函数 f ，其实质等同于将 f 重新定义为下一个残差单元的激活前处理。图5对此进行了直观展示。

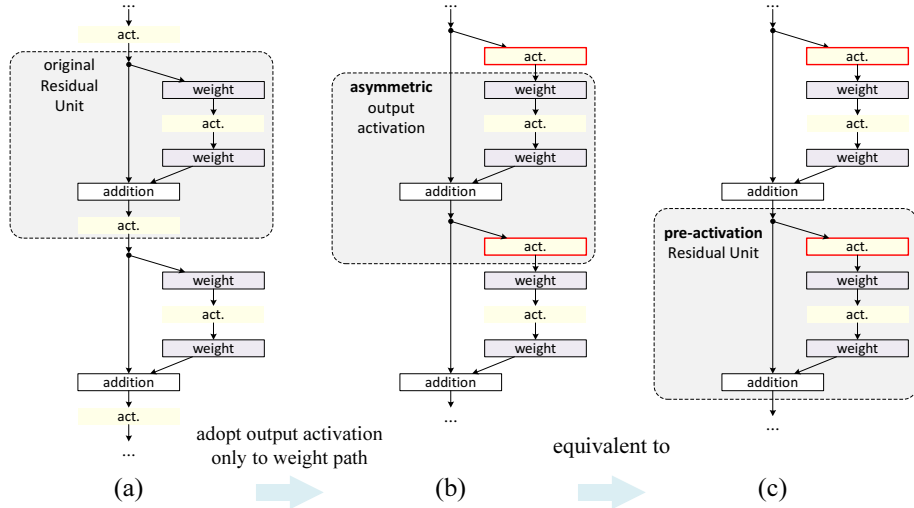


Figure 5. Using asymmetric after-addition activation is equivalent to constructing a *pre-activation* Residual Unit.

Table 3. Classification error (%) on the CIFAR-10/100 test set using the original Residual Units and our pre-activation Residual Units.

dataset	network	baseline unit	pre-activation unit
CIFAR-10	ResNet-110 (1layer skip)	9.90	<u>8.91</u>
	ResNet-110	6.61	<u>6.37</u>
	ResNet-164	5.93	<u>5.46</u>
	ResNet-1001	7.61	<u>4.92</u>
CIFAR-100	ResNet-164	25.16	<u>24.33</u>
	ResNet-1001	27.82	<u>22.71</u>

The distinction between post-activation/pre-activation is caused by the presence of the element-wise *addition*. For a plain network that has N layers, there are $N - 1$ activations (BN/ReLU), and it does not matter whether we think of them as post- or pre-activations. But for branched layers merged by addition, the position of activation matters.

We experiment with two such designs: (i) ReLU-only pre-activation (Fig. 4(d)), and (ii) full pre-activation (Fig. 4(e)) where BN and ReLU are both adopted before weight layers. Table 2 shows that the ReLU-only pre-activation performs very similar to the baseline on ResNet-110/164. This ReLU layer is not used in conjunction with a BN layer, and may not enjoy the benefits of BN [8].

Somehow surprisingly, when BN and ReLU are both used as pre-activation, the results are improved by healthy margins (Table 2 and Table 3). In Table 3 we report results using various architectures: (i) ResNet-110, (ii) ResNet-164, (iii) a 110-layer ResNet architecture in which each shortcut skips only 1 layer (*i.e.*,

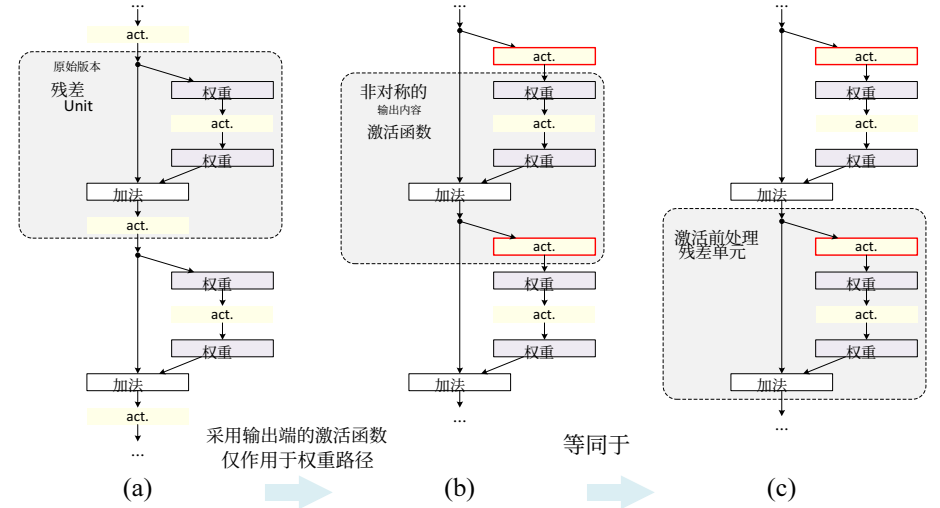


图5. 使用非对称的加法后的激活函数，等同于构建一个残差单元。

表3. 使用传统残差单元与我们所设计的预激活残差单元在CIFAR-10/100测试集上的分类误差百分比。

数据集	网络	基准方案单元	预激活单元
CIFAR-10数据集	ResNet-110网络 (1层跳跃连接)	9.90	<u>8.91</u>
	ResNet-110网络	6.61	<u>6.37</u>
	ResNet-164网络	5.93	<u>5.46</u>
	ResNet-1001网络	7.61	<u>4.92</u>
CIFAR-100数据集	ResNet-164网络	25.16	<u>24.33</u>
	ResNet-1001网络	27.82	<u>22.71</u>

激活后处理与激活前处理之间的区别，源于元素级加法的存在。在拥有 N 层的普通网络中，存在 $N - 1$ 个激活函数（即批归一化与ReLU函数），无论我们将它们视为激活后处理还是激活前处理并无实质差异。但对于通过加法连接起来的分支层而言，激活函数所处的位置则至关重要。

我们采用了两种设计方案进行实验：(i) 仅使用ReLU的激活前处理（图4(d)），以及(ii) 完全预激活处理（图4(e)），后者在权重层之前同时应用了批归一化与ReLU函数。表2显示，在ResNet-110/164网络上，仅使用ReLU的激活前处理的效果与基准方案极为接近。由于这种ReLU层并未与批归一化层结合使用，因此可能无法享受到批归一化的优势 [8]。

令人惊讶的是，当同时将批归一化和ReLU用作激活前处理时，结果竟会展现出显著的优势（见表2和表3）。在表3中，我们展示了多种不同架构下的实验结果：(i) ResNet-110网络，(ii) ResNet-164网络，(iii) 一种仅有110层的ResNet架构，其每条捷径仅跳过1层（即），

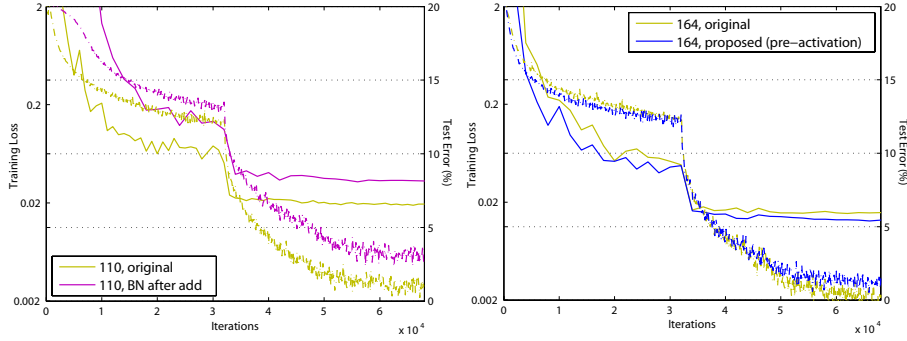


Figure 6. Training curves on CIFAR-10. **Left:** BN after addition (Fig. 4(b)) using ResNet-110. **Right:** pre-activation unit (Fig. 4(e)) on ResNet-164. Solid lines denote test error, and dashed lines denote training loss.

a Residual Unit has only 1 layer), denoted as “ResNet-110(1layer)”, and (iv) a 1001-layer bottleneck architecture that has 333 Residual Units (111 on each feature map size), denoted as “ResNet-1001”. We also experiment on CIFAR-100. Table 3 shows that our “pre-activation” models are consistently better than the baseline counterparts. We analyze these results in the following.

4.2 Analysis

We find the impact of pre-activation is twofold. First, the optimization is further eased (comparing with the baseline ResNet) because f is an identity mapping. Second, using BN as pre-activation improves regularization of the models.

Ease of optimization. This effect is particularly obvious when training the 1001-layer ResNet. Fig. 1 shows the curves. Using the original design in [1], the training error is reduced very slowly at the beginning of training. For $f = \text{ReLU}$, the signal is impacted if it is negative, and when there are many Residual Units, this effect becomes prominent and Eqn.(3) (so Eqn.(5)) is not a good approximation. On the other hand, when f is an identity mapping, the signal can be propagated directly between any two units. Our 1001-layer network reduces the training loss very quickly (Fig. 1). It also achieves the lowest loss among all models we investigated, suggesting the success of optimization.

We also find that the impact of $f = \text{ReLU}$ is not severe when the ResNet has fewer layers (*e.g.*, 164 in Fig. 6(right)). The training curve seems to suffer a little bit at the beginning of training, but goes into a healthy status soon. By monitoring the responses we observe that this is because after some training, the weights are adjusted into a status such that $\mathbf{y}_l$ in Eqn.(1) is more frequently above zero and f does not truncate it ($\mathbf{x}_l$ is always non-negative due to the previous ReLU, so $\mathbf{y}_l$ is below zero only when the magnitude of $\mathcal{F}$ is very negative). The truncation, however, is more frequent when there are 1000 layers.

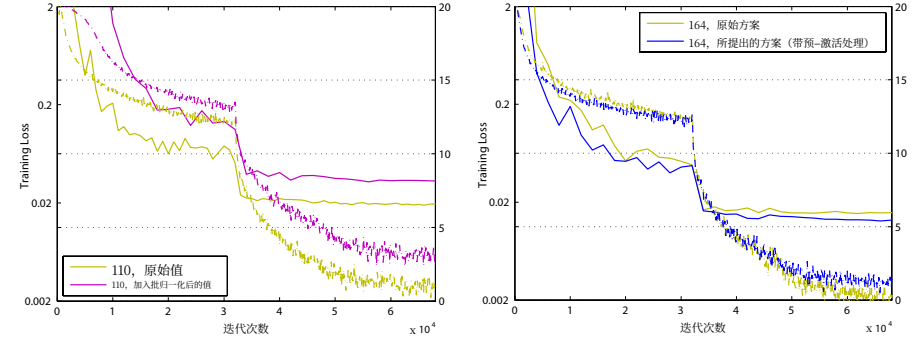


图6. CIFAR-10数据集上的训练曲线。左侧为采用ResNet-110的加法后归一化方案（图4(b)）；右侧为在ResNet-164上应用的预激活单元方案（图4(e)）。实线代表测试误差，虚线代表训练损失。

该架构中的每个残差单元也仅包含1层，因此被标记为“ResNet-110(1layer)”；此外还有 (iv) 一种拥有1001层且包含333个残差单元的瓶颈型架构，每个特征图尺寸上对应111个残差单元，该架构被标记为“ResNet-1001”。我们还在CIFAR-100数据集上进行了实验。表3显示，采用这种“激活前处理”方法的模型表现始终优于基准方案。接下来我们将对这些实验结果进行详细分析。

4.2 分析

我们发现预激活处理的作用主要有两方面。首先，由于 f 属于恒等映射，优化过程会变得更加轻松（相比基于基准ResNet的方案）。其次，将批归一化用作预激活处理有助于提升模型的正则化效果。

优化的便捷性。这一效应在训练[1], 1001层的ResNet时尤为明显。图1展示了相应的曲线：采用[1]中的原始设计时，训练初期的训练误差下降速度十分缓慢。对于 $f = \text{ReLU}$ 激活函数而言，若输入信号为负值就会受到影响；而当残差单元数量较多时，这种影响会愈发显著，此时方程(3)（进而方程(5)）也就不再是一个恰当的近似公式。相反，当 f 为恒等映射时，信号便能直接在任意两个单元之间传递。我们所研究的1001层网络能够非常迅速地降低训练损失（见图1），并且在其所有对比模型中取得了最低的损失值，这充分证明了该优化方法的有效性。

我们还将发现，当ResNet的层数较少时（例如，图6右侧的164层）， $f = \text{ReLU}$ 带来的影响并不严重。训练曲线在训练初期似乎会有些波动，但很快就会恢复稳定状态。通过监测相关响应，我们发现这是因为经过一定程度的训练后，权重会被调整到这样一种状态：方程(1)中的 $\mathbf{y}_l$ 更频繁地处于正值，同时 f 也不会对其造成截断现象（由于之前的ReLU作用， $\mathbf{x}_l$ 始终为非负数，因此只有当 $\mathcal{F}$ 的绝对值非常负时， $\mathbf{y}_l$ 才会出现负值）。不过，当层数达到1000层时，这种截断现象出现的频率会更高。

Table 4. Comparisons with state-of-the-art methods on CIFAR-10 and CIFAR-100 using “*moderate data augmentation*” (flip/translation), except for ELU [12] with no augmentation. Better results of [13,14] have been reported using stronger data augmentation and ensembling. For the ResNets we also report the number of parameters. Our results are the median of 5 runs with mean $\pm$ std in the brackets. All ResNets results are obtained with a mini-batch size of 128 except [†] with a mini-batch size of 64 (code available at <https://github.com/KaimingHe/resnet-1k-layers>).

CIFAR-10	error (%)	CIFAR-100	error (%)
NIN [15]	8.81	NIN [15]	35.68
DSN [16]	8.22	DSN [16]	34.57
FitNet [17]	8.39	FitNet [17]	35.04
Highway [7]	7.72	Highway [7]	32.39
All-CNN [14]	7.25	All-CNN [14]	33.71
ELU [12]	6.55	ELU [12]	24.28
FitResNet, LSUV [18]	5.84	FitNet, LSUV [18]	27.66
ResNet-110 [1] (1.7M)	6.61	ResNet-164 [1] (1.7M)	25.16
ResNet-1202 [1] (19.4M)	7.93	ResNet-1001 [1] (10.2M)	27.82
ResNet-164 [ours] (1.7M)	5.46	ResNet-164 [ours] (1.7M)	24.33
ResNet-1001 [ours] (10.2M)	4.92 (4.89 $\pm$ 0.14)	ResNet-1001 [ours] (10.2M)	22.71 (22.68 $\pm$ 0.22)
ResNet-1001 [ours] (10.2M) [†]	4.62 (4.69 $\pm$ 0.20)		

Reducing overfitting. Another impact of using the proposed pre-activation unit is on regularization, as shown in Fig. 6 (right). The pre-activation version reaches slightly higher training loss at convergence, but produces lower test error. This phenomenon is observed on ResNet-110, ResNet-110(1-layer), and ResNet-164 on both CIFAR-10 and 100. This is presumably caused by BN’s regularization effect [8]. In the original Residual Unit (Fig. 4(a)), although the BN normalizes the signal, this is soon added to the shortcut and thus the merged signal is not normalized. This unnormalized signal is then used as the input of the next weight layer. On the contrary, in our pre-activation version, the inputs to all weight layers have been normalized.

5 Results

Comparisons on CIFAR-10/100. Table 4 compares the state-of-the-art methods on CIFAR-10/100, where we achieve competitive results. We note that we do not specially tailor the network width or filter sizes, nor use regularization techniques (such as dropout) which are very effective for these small datasets. We obtain these results via a simple but essential concept — going deeper. These results demonstrate the potential of *pushing the limits of depth*.

Comparisons on ImageNet. Next we report experimental results on the 1000-class ImageNet dataset [3]. We have done preliminary experiments using the skip connections studied in Fig. 2 & 3 on ImageNet with ResNet-101 [1], and observed similar optimization difficulties. The training error of these non-identity shortcut networks is obviously higher than the original ResNet at the first learning rate

表4.该表展示了在CIFAR-10和CIFAR-100数据集上，采用“*moderate data augmentation*”（图像翻转/平移）技术对比最先进的算法的测试结果，其中不包括未进行任何增强处理的ELU 算法 [12]。通过使用更强大的数据增强手段以及集成学习方法，已有研究获得了更好的性能表现。对于ResNet系列网络，我们还列出了其网络参数数量。此处呈现的结果为5次实验的中位数，均值及标准差则标注在括号内。所有ResNet相关的实验都是以128为小批量大小进行的，[†] 仅有部分实验采用64作为小批量大小（相关代码可见于<https://github.com/KaimingHe/resnet-1k-layers>）。

CIFAR-10数据集	误差百分比	CIFAR-100数据集	误差百分比
NIN [15]	8.81	NIN [15]	35.68
DSN模型 [16]	8.22	DSN [16]	34.57
FitNet模型 [17]	8.39	FitNet [17]	35.04
Highway模型 [7]	7.72	Highway [7]	32.39
All-CNN [14]	7.25	All-CNN模型 [14]	33.71
ELU [12]	6.55	ELU激活函数 [12]	24.28
FitResNet与LSUV [18]	5.84	FitNet, LSUV [18]	27.66
ResNet-110 [1] (170万参数)	6.61	ResNet-164 [1] (170万参数)	25.16
ResNet-1202 [1] (1940万参数)	7.93	ResNet-1001 [1] (1020万参数)	27.82
ResNet-164 [ours] (170万参数)	5.46	ResNet-164 [ours] (170万参数)	24.33
ResNet-1001 [ours] (1020万参数)	4.92 (4.92)	ResNet-1001 [ours] (1020万参数)	22.71 (22.68 $\pm$ 0.22)
ResNet-1001 [ours] (1020万参数) [†]	4.62 (4.92)		

降低过拟合现象。如图6（右侧）所示，采用所提出的预激活单元的另一个影响体现在正则化方面。在收敛过程中，预激活版本虽然会带来略高的训练损失，但能够降低测试误差。这一现象在CIFAR-10和CIFAR-100数据集上的ResNet-110、ResNet-110（1层版）以及ResNet-164网络中均有出现。这很可能是由于批归一化的正则化作用 [8]所致。在原有的残差单元结构（图4(a)）中，尽管批归一化会对信号进行标准化处理，但这些标准化后的信号很快就会被加入快捷路径中，因此合并后的信号并未真正实现标准化。而后这个未标准化的信号又会作为下一权重层的输入。与之相反，在我们设计的预激活版本中，所有权重层的输入都经过了标准化处理。

5个研究结果

CIFAR-10/100数据集上的对比分析。表4展示了在CIFAR-10/100数据集上各种最先进的算法表现，而我们在其中取得了具有竞争力的优异成绩。需要指出的是，我们并未特意调整网络的宽度或滤波器的尺寸，也没有使用那些对这些小型数据集极为有效的正则化技术（如丢弃机制）。我们之所以能取得这样的成果，靠的是一个简单却至关重要的方法——增加网络的深度。这些结果充分体现了不断拓展深度极限的巨大潜力。

在ImageNet数据集上的对比分析。接下来，我们将汇报在包含1000个类别的ImageNet数据集 [3]上进行的实验结果。我们曾使用图2和图3中研究的跳过连接机制，在ImageNet数据集上结合ResNet-101 [1]，进行了初步实验，发现同样存在类似的优化难题。在这些非恒等映射结构的快捷网络中，以初始学习率训练时的训练误差明显高于原始ResNet网络。

Table 5. Comparisons of single-crop error on the ILSVRC 2012 validation set. All ResNets are trained using the same hyper-parameters and implementations as [1]. Our Residual Units are the full pre-activation version (Fig. 4(e)). [†]: code/model available at <https://github.com/facebook/fb.resnet.torch/tree/master/pretrained>, using scale and aspect ratio augmentation in [20].

method	augmentation	train crop	test crop	top-1	top-5
ResNet-152, original Residual Unit [1]	scale	224×224	224×224	23.0	6.7
ResNet-152, original Residual Unit [1]	scale	224×224	320×320	21.3	5.5
ResNet-152, pre-act Residual Unit	scale	224×224	320×320	21.1	5.5
ResNet-200, original Residual Unit [1]	scale	224×224	320×320	21.8	6.0
ResNet-200, pre-act Residual Unit	scale	224×224	320×320	20.7	5.3
ResNet-200, pre-act Residual Unit	scale+asp ratio	224×224	320×320	20.1[†]	4.8[†]
Inception v3 [19]	scale+asp ratio	299×299	299×299	21.2	5.6

(similar to Fig. 3), and we decided to halt training due to limited resources. But we did finish a “BN after addition” version (Fig. 4(b)) of ResNet-101 on ImageNet and observed higher training loss and validation error. This model’s single-crop (224×224) validation error is 24.6%/7.5%, *vs.* the original ResNet-101’s 23.6%/7.1%. This is in line with the results on CIFAR in Fig. 6 (left).

Table 5 shows the results of ResNet-152 [1] and ResNet-200³, all trained from scratch. We notice that the original ResNet paper [1] trained the models using scale jittering with shorter side $s \in [256, 480]$, and so the test of a 224×224 crop on $s = 256$ (as did in [1]) is negatively biased. Instead, we test a single 320×320 crop from $s = 320$, for all original and our ResNets. Even though the ResNets are trained on smaller crops, they can be easily tested on larger crops because the ResNets are fully convolutional by design. This size is also close to 299×299 used by Inception v3 [19], allowing a fairer comparison.

The original ResNet-152 [1] has top-1 error of 21.3% on a 320×320 crop, and our pre-activation counterpart has 21.1%. The gain is not big on ResNet-152 because this model has not shown severe generalization difficulties. However, the original ResNet-200 has an error rate of 21.8%, higher than the baseline ResNet-152. But we find that the original ResNet-200 has *lower* training error than ResNet-152, suggesting that it suffers from overfitting.

Our pre-activation ResNet-200 has an error rate of 20.7%, which is **1.1%** lower than the baseline ResNet-200 and also lower than the two versions of ResNet-152. When using the scale and aspect ratio augmentation of [20,19], our ResNet-200 has a result better than Inception v3 [19] (Table 5). Concurrent with our work, an Inception-ResNet-v2 model [21] achieves a single-crop result of 19.9%/4.9%. We expect our observations and the proposed Residual Unit will help this type and generally other types of ResNets.

Computational Cost. Our models’ computational complexity is linear on

³ The ResNet-200 has 16 more 3-layer bottleneck Residual Units than ResNet-152, which are added on the feature map of 28×28.

表5. ILSVRC 2012数据集验证集中各单张图像误差的对比情况。所有ResNet网络均采用与 [1]中相同的超参数及实现方案进行训练。此外，我们所使用的残差单元为采用完全预激活版本的恒等映射函数（见图4(e)）。[†]：相关代码/模型可在

<https://github.com/facebook/fb.resnet.torch/tree/master/pretrained>获取，该实现方案还运用了缩放与宽高比增强技术 [20]。

方法	增强处理	训练裁剪区域	测试裁剪区域	前1名精度	前5名精度
ResNet-152网络，原始的残差单元 [1]	缩放	224×224	224×224	23.0	6.7
ResNet-152网络，原始的残差单元 [1]	缩放	224×224	320×320	21.3	5.5
ResNet-152网络， 预激活 残差单元	缩放比例	224×224	320×320	21.1	5.5
ResNet-200，原始的残差单元 [1]	缩放	224×224	320×320	21.8	6.0
ResNet-200， pre-act 残差单元	缩放比例	224×224	320×320	20.7	5.3
ResNet-200， pre-act 残差单元	比例值为scale+asp ratio	224×224	320×320	20.1[†]	4.8[†]
Inception v3 [19]	比例值为scale+asp ratio	299×299	299×299	21.2	5.6

(情况与图3类似)，由于资源有限，我们不得不终止训练。不过我们还是完成了在ImageNet数据集上针对ResNet-101的“加法后归一化”版本（图4(b)）的测试，结果发现该版本的训练损失和验证误差均更高。该模型在单张图像输入条件下的(224×224)验证误差为24.6%/7.5%，而原始ResNet-101的对应数值则为23.6%/7.1%。这一结果与图6（左侧）中CIFAR数据集上的实验结果一致。

表5展示了从零开始训练的ResNet-152 [1] 与ResNet-200³，的实验结果。我们注意到，原始的ResNet论文 [1] 采用尺度抖动技术，并使用较短的边长 $s \in [256, 480]$ 来训练模型，因此在对 $s = 256$ 进行 224×224 尺寸裁剪后的测试结果（与 [1]中的做法一致）存在负面偏差。为此，我们对所有原始模型以及我们的ResNet系列网络都采用了来自 $s = 320$ 的单一 320×320 尺寸裁剪进行测试。尽管这些ResNet网络是在较小的裁剪区域上训练的，但由于其设计上即为全卷积结构，因此能够轻松地更大的裁剪区域上进行测试。此外，这一尺寸也与 Inception v3 [19]，所使用的尺寸相近，从而使得对比更加公平。

原始的ResNet-152 [1] 在 320×320 裁剪数据集上的前1名精度误差为21.3%，而我们采用激活前处理方法的对应模型的误差则为21.1%。由于ResNet-152并未表现出严重的泛化能力问题，因此其误差改善幅度并不显著。相比之下，原始的ResNet-200的误差率高达21.8%，超过了作为基准方案的ResNet-152。不过我们发现，原始ResNet-200的训练误差反而低于ResNet-152，这一现象表明该模型存在过拟合问题。

我们经过激活前处理处理的ResNet-200的误差率为20.7%，这一数值比作为基准方案的ResNet-200低**1.1%**，同时也低于两种版本的ResNet-152。在应用尺寸缩放和宽高比增强技术后，我们的ResNet-200的测试效果甚至优于Inception v3 [19]（见表5）。与我们的研究同时进行的另一项工作中，一种名为Inception-ResNet-v2的模型 [21] 在单次裁剪数据集上的测试成绩为19.9%/4.9%。我们相信，我们的研究结果以及所提出的残差单元将有助于提升这类以及其它类型的ResNet网络的性能。

计算成本。我们模型的计算复杂度为线性级。

³ ResNet-200模型相比ResNet-152网络多出了16个三层瓶颈残差单元，这些单元被添加在 28×28的特征图上。

depth (so a 1001-layer net is $\sim 10\times$ complex of a 100-layer net). On CIFAR, ResNet-1001 takes about 27 hours to train on 2 GPUs; on ImageNet, ResNet-200 takes about 3 weeks to train on 8 GPUs (on par with VGG nets [22]).

6 Conclusions

This paper investigates the propagation formulations behind the connection mechanisms of deep residual networks. Our derivations imply that identity shortcut connections and identity after-addition activation are essential for making information propagation smooth. Ablation experiments demonstrate phenomena that are consistent with our derivations. We also present 1000-layer deep networks that can be easily trained and achieve improved accuracy.

Appendix: Implementation Details The implementation details and hyper-parameters are the same as those in [1]. On CIFAR we use only the translation and flipping augmentation in [1] for training. The learning rate starts from 0.1, and is divided by 10 at 32k and 48k iterations. Following [1], for all CIFAR experiments we warm up the training by using a smaller learning rate of 0.01 at the beginning 400 iterations and go back to 0.1 after that, although we remark that this is not necessary for our proposed Residual Unit. The mini-batch size is 128 on 2 GPUs (64 each), the weight decay is 0.0001, the momentum is 0.9, and the weights are initialized as in [23].

On ImageNet, we train the models using the same data augmentation as in [1]. The learning rate starts from 0.1 (no warming up), and is divided by 10 at 30 and 60 epochs. The mini-batch size is 256 on 8 GPUs (32 each). The weight decay, momentum, and weight initialization are the same as above.

When using the pre-activation Residual Units (Fig. 4(d)(e) and Fig. 5), we pay special attention to the first and the last Residual Units of the entire network. For the first Residual Unit (that follows a stand-alone convolutional layer, conv_1), we adopt the first activation right after conv_1 and before splitting into two paths; for the last Residual Unit (followed by average pooling and a fully-connected classifier), we adopt an extra activation right after its element-wise addition. These two special cases are the natural outcome when we obtain the pre-activation network via the modification procedure as shown in Fig. 5.

The bottleneck Residual Units (for ResNet-164/1001 on CIFAR) are constructed following [1]. For example, a $\begin{bmatrix} 3\times 3, 16 \\ 3\times 3, 16 \end{bmatrix}$ unit in ResNet-110 is replaced with a $\begin{bmatrix} 1\times 1, 16 \\ 3\times 3, 16 \\ 1\times 1, 64 \end{bmatrix}$ unit in ResNet-164, both of which have roughly the same number of parameters. For the bottleneck ResNets, when reducing the feature map size we use projection shortcuts [1] for increasing dimensions, and when pre-activation is used, these projection shortcuts are also with pre-activation.

深度（因此，1001层网络相比100层网络要 $\sim 10\times$ 复杂得多）。在CIFAR数据集上，使用2个GPU训练ResNet-1001大约需要27小时；而在ImageNet数据集上，使用8个GPU训练ResNet-200则需要大约3周时间（其训练效率与VGGnet系列模型 [22]相当）。

6 结论

本文研究了深度残差网络中连接机制背后的信息传播公式。我们的推导表明，恒等映射快捷连接以及加法后恒等激活函数对于实现顺畅的信息传播至关重要。消融实验得出的结果与我们的推导一致。此外，我们还提出了层级数为1000的深度网络，这类网络不仅易于训练，还能提升准确率。

附录：实现细节其实现细节与超参数均与 [1]中的相同。在CIFAR数据集上的训练过程中，我们仅采用 [1] 中规定的图像平移和翻转增强技术。学习率初始值为0.1，在迭代次数达到32k和48k时分别除以10。遵循 [1]，中的方法，我们在前400次迭代时使用0.01这一较低的学习率进行热身训练，之后再恢复到0.1的学习率；不过需要说明的是，对于我们提出的残差单元而言，这一步并非必需。该模型的小批量大小为128，对应2个GPU（每个GPU处理64个样本），权重衰减系数为0.0001，动量为0.9，而网络权重的初始化方式则与 [23]中的规定一致。

在ImageNet数据集上训练模型时，我们采用了与[1]中相同的策略来进行数据增强。学习率初始值为0.1，且不进行预热处理；在训练30个周期和60个周期时，学习率分别除以10。此时使用8台GPU（每台分配32个批次）进行处理，小批量大小为256。此外，权重衰减、动量以及参数初始化的方式都与前述设置一致。

当使用预激活残差单元（见图4(d)(e)和图5）时，我们需要特别关注整个网络中的第一个和最后一个残差单元。对于紧跟在独立卷积层 conv_1 之后的第一个残差单元，我们会采用 conv_1 之后、路径分裂之前处的第一个激活函数；而对于紧接着平均池化和全连接分类器之后的最后一个残差单元，则会在其执行逐元素相加操作之后再加入一个额外的激活函数。这两种特殊处理方式，正是通过图5所示的修改流程构建预激活网络时自然而然会出现的结果。

CIFAR数据集上的ResNet-164/1001版本所采用的瓶颈残差单元是依据 [1] 的构想构建的。例如，ResNet-110中的 $\begin{bmatrix} 3\times 3, 16 \\ 3\times 3, 16 \end{bmatrix}$ 类型单元会被替换为ResNet-164中的 $\begin{bmatrix} 1\times 1, 16 \\ 3\times 3, 16 \\ 1\times 1, 64 \end{bmatrix}$ 类型单元，这两种单元的网络参数数量大致相同。对于这类瓶颈型ResNet网络，在缩小特征图尺寸时，会使用 [1] 这种投影快捷路径来提升维度；而当采用激活前处理机制时，这些投影快捷路径也会同时集成激活前处理功能。

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